

• Power of Compounding:

- Q. Sachin's father invested ₹10,000 in year 2001 in bank deposit scheme with an avg. interest rate of 6.5%. ~~calculate a total am~~ Sachin's father withdraws a FD in year ²⁰¹⁰ ~~2021~~. calculate the total amount by simple and compound interest rate.

⇒ year	Return	Return + Principal	
⇒ 2001	650	10,650	
2002	650	11,300	
2003	650	11,950	
2004	650	12,600	
2005	650	13,250	
2006	650	13,900	
2007	650	14,550	
2008	650	15,200	
2009	650	15,850	
2010	650	16,500	→ simple interest
⇒ 2001	650	10,650	
2002	692.25	11,342.25	
2003	737.24	12,079.49	
2004	785.16	12,864.65	
2005	836.2	13,700.85	
2006	890.55	14,591.4	
2007	948.44	15,539.84	
2008	1010.08	15,549.92	
2009	1075.74	16,625.66	
2010	1080.66	17,706.32	→ compound interest

• Rule of 72:

- Q. Sachin invested ₹1,00,000 in BOI at interest rate of 7%. Sachin want to get double the amount, how much time is required for it.
- ⇒ $72 \div 7 = 10.2857$ years.

→ It will take Sachin 10.28 years to double his amount of investment.

• Simple Interest →

$$S.I. = (P \times R \times T) / 100$$

where P → principal

R → Rate of interest in % per annum.

T → Time

- Q. $S.I. = ₹ 4016.25$ $R = 9\%$ per annum
 $T = 5$ years. $P = ?$

$$\Rightarrow S.I. = \frac{4016.25 \times 9 \times 5}{100}$$

$$= 1807.3125$$

$$\text{Total} : 4016.25 + 1807.3125 = ₹ 5823.5625$$

$$4016.25 = \frac{P \times 9 \times 5}{100}$$

$$P = \frac{401625}{45}$$

$$P = ₹ 8925.$$

Q. $P = ₹ 8000$ $T = 15$ month $\approx$

$R = 6$ paise per rupee per month.

S.I. = ?

$\Rightarrow 6$ paise / 1 ₹ rup.

$$\frac{6}{100} = 6\%$$

$$S.I. = \frac{P \times R \times T}{100} = \frac{8000 \times 6 \times 15}{100}$$

$$S.I. = ₹ 7200$$

Prin

$$\text{Principle} + \text{Return} = 8000 + 7200 = ₹ 15,200$$

• Compound Interest $\Rightarrow$

$$C.I. = P \left(1 + \frac{r}{n} \right)^{nt} - P$$

where, $P \rightarrow$ principle amount

$n \rightarrow$ compounding freq. per annum.

$r \rightarrow$ rate of interest

$t \rightarrow$ Time.

Q. Find amount and C.I. on ₹ 12,000 in 3 years at 5% compounded annually.

$$\Rightarrow C.I. = 12000 \left(1 + \frac{5}{100} \right)^{1 \times 3} - 12,000$$

$$= 12,000 \left(\frac{105}{100} \right)^3 - 12,000$$

$$= 12,000 (1 + 0.05)^3 - 12,000$$

$$C.I. = \cancel{1891} \cdot 1891.5$$

$$A = 12,000 + 1891.5 = ₹ 13,891.5$$

Q. Sum of ₹ 50,000 is borrowed at rate of interest is 10% per annum, what is CI for 5 years.

$$\Rightarrow C.I. = 50,000 (1 + 0.1)^5 - 50,000 = ₹ 30,525.5$$

$$A = ₹ 80,525.5$$